

Content

Learning objectives and content

Objectives

After completing the course, the student

- * knows and can work with a mathematical representation of system behaviour (labelled transition systems modulo a behavioural equivalence);
- * can specify behaviour in a language including facilities for non-determinism, parallelism, and abstraction, and can specify simple requirements of behaviour in a modal logic with fixed points;
- * knows and understands the semantics of both the behaviour specification language and the modal logic;
- * is familiar with several approaches to determining the correctness of behaviour;
- * can reason about the correctness of system behaviour, with and without using tool support;
- * is familiar with the algorithms and mathematical theory underlying verification of system behaviour.

Content

The course will address the following topics:

- * labelled transition systems as the primary means to mathematically represent system behaviour;
- * refinement preorders, trace equivalence, strong bisimilarity, and observational equivalence on labelled transition systems;
- * algorithms to compute whether states in a transition system are bisimilar
- * mathematical theory of fixed points;
- * syntax and semantics of CCS (Calculus of Communicating Systems);
- * syntax and semantics of HML (Hennessy-Milner Logic) with recursion;
- * abstraction and refinement.

There will be a practical assignment in which simple system behaviour must be modelled and analysed using tool support.

2026

Process Theory (2ITA0)

Practical information

Course module

2ITA0

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Advanced

Language of instruction

English

Offered by

Mathematics and Computer Science - Computer Science domain

General course information

Assessment: Students can decide, during the first week of the course, whether they want to participate in optional (graded) homework assignments. Then: A mandatory practical assignment weights 20% of the final grade; Homework assignments weight 30% of the final grade, only for students participating to the homework assignments; A written test weights 50% for students participating in the homework assignments, and 80% for students who opted out from those. The grade of the written exam needs to be at least 5.0 to pass the course.

Short promotional description of the course

Modern computing systems typically consist of many components running in parallel, and interacting with each other and their environment. They are not designed to compute a result, but, rather, to execute a process and thus exhibit behavior. For a correct design of a modern computing system, it is therefore essential to be able to specify its intended behavior, at different levels of abstraction, and to analyze and reason about this behavior. This course offers a formal approach to specifying and reasoning about system behavior, and it introduces the algorithmics for analyzing system behavior.

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3

Timeslot(s)

E - Mo 9-10, Tu 5-8, Th 1-4

Enrolment period

15 November 2026 until 3 January 2027 23:59

Unenrolment

until 1 February 2027 23:59

Entrance requirements

Assumed previous knowledge

Elementary notions of (predicate) logic and set theory and proving skills as taught in, e.g., Logic and Set Theory (2IT60) and Introduction to Discrete Structures (2IT80), or Set Theory and Algebra (2WF40). Students are expected to have had some prior exposure to formal topics in computer science, e.g., because they have done Automata, language theory and complexity (2IT90)), Automata and Formal Languages (2IRR90) or Software Specification (2IX20).

Entrance requirements course enrolment

You must meet the following requirements

- Registered for a degree programme other than
 - HBO-TOP Applied Physics, Pre-Master, Fulltime

Instructional modes

Instructional modes

- Labsession

Remark

- Lecture

Remark

- Tutorial
Remark

Tests

Tests

- Written examination
Final examination

Test weight
100

Minimum grade
6.0

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Written examination

- Block 3
Enrolment period
15 November 2026 until 21 March 2027 23:59

Unenrolment
until 21 March 2027

Test date
6 April 2027

- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Unenrolment
until 6 June 2027

Test date
22 June 2027

Lecturer(s)

Co-lecturer

- V. Castiglioni

Contactperson for the course

- prof.dr.ir. T.A.C. Willemse

Responsible lecturer

- prof.dr.ir. T.A.C. Willemse

Subject matter expert

- prof.dr.ir. T.A.C. Willemse
- V. Castiglioni

Materials

Recommended material

- Reactive Systems: Modelling, Specification and Verification by Luca Aceto, Anna Ingolfsdottir, Kim G. Larsen, Jiri Srba. Cambridge University Press, July 2007. (only Part I of the book will be used)
- Handouts
- Slides

Do you want to know more?

prof.dr.ir. T.A.C. Willemse can tell you more about this course.

Send an e-mail to T.A.C.Willemse@tue.nl